§7.1 Diagonalization of symmetric matrices

- In chapter 7 we will only working with <u>real</u> matrices, i.e. we will <u>not</u> work over $\mathbb{C}$.

- In chapter 7 we will only work with square i.e. $n \times n$ matrices. These matrices lie in $M_{n \times n}$

We are interested in two kinds of square matrices: <u>symmetric matrices</u> and <u>orthogonal matrices</u>

Def. A square matrix $A \in M_{n \times n}$ is <u>symmetric</u> if $A^T = A$. i.e. if

$$A = \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \vdots \\ a_{n1} & \cdots & a_{nn} \end{bmatrix} \quad \text{then}$$

$$a_{jk} = a_{kj}.$$

Examples

Let $A = \begin{bmatrix} 1 & 2 \\ 2 & 7 \end{bmatrix}$. Then $A^T = \begin{bmatrix} 1 & 2 \\ 2 & 7 \end{bmatrix}$

$= A$. is symmetric

By contrast $\begin{bmatrix} 1 & 2 \\ 3 & 7 \end{bmatrix}^T = \begin{bmatrix} 1 & 3 \\ 2 & 7 \end{bmatrix} \neq \begin{bmatrix} 1 & 2 \\ 3 & 7 \end{bmatrix}$

$\Rightarrow \begin{bmatrix} 1 & 2 \\ 3 & 7 \end{bmatrix}$ is <u>not</u> symmetric.

Also $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ is <u>not</u> symmetric since not square.

Take $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{bmatrix}$. Then $A^T = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{bmatrix}$

$= A.$

Recall that the jk entry of A^T is (by def'n) the kj entry of A.

Example

Suppose $B \in M_{n \times m}$ then $BB^T \in M_{n \times n}$.

and $(BB^T)^T = (B^T)^T B^T = BB^T$

Therefore BB^T is a symmetric matrix.

Remarks - Transpose does not affect diagonal entries of a matrix. So symmetric matrices have no restrictions on diagonal entries.

- Transpose converts the columns of a matrix into rows (an vice versa)

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}^T = \begin{bmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{bmatrix}$$

i.e. $$\begin{bmatrix} u_1 & u_2 & \cdots & u_n \end{bmatrix}^T = \begin{bmatrix} u_1^T \\ \vdots \\ u_n^T \end{bmatrix}$$

where $u_1, \ldots, u_n$ are columns of the matrix.

Def A square matrix $P \in M_{n \times n}$ is <u>orthogonal</u> if $P^T = P^{-1}$.

(in particular P is invertible)

i.e. $P^T P = I = P P^T$

identity matrix.

(Enough to check $P^T P = I$)

Example.

Let $P = \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}$.

Let's compute $P^T P = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ -1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}$

$$= \begin{bmatrix} (1/\sqrt{2})^2 + (1/\sqrt{2})^2 & -(1/\sqrt{2})^2 + (1/\sqrt{2})^2 \\ -(1/\sqrt{2})^2 + (1/\sqrt{2})^2 & (1/\sqrt{2})^2 + 1/\sqrt{2})^2 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I.$$

We may conclude that P is orthogonal.

Let $P = [u_1 \ u_2]$ where $u_1 = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}$, $u_2 = \begin{bmatrix} -1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}$

are the columns of P.

Then $P^T = \begin{bmatrix} u_1^T \\ u_2^T \end{bmatrix}$ and

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = P^T P = \begin{bmatrix} u_1^T u_1 & u_1^T u_2 \\ u_2^T u_1 & u_2^T u_2 \end{bmatrix}$$

$$= \begin{bmatrix} \langle u_1, u_1 \rangle & \langle u_1, u_2 \rangle \\ \langle u_2, u_1 \rangle & \langle u_2, u_2 \rangle \end{bmatrix}$$

$$\Leftrightarrow \quad \begin{aligned} 1 &= \langle u_1, u_1 \rangle \\ 0 &= \langle u_1, u_2 \rangle \\ 0 &= \langle u_2, u_1 \rangle \\ 1 &= \langle u_2, u_2 \rangle \end{aligned}$$

$\Longrightarrow$ $\{u_1, u_2\}$ is an orthonormal basis!!

All we used to get this conclusion was that $P^T P = I$ (i.e. P is orthogonal).

Theorem

Let $u, v \in \mathbb{R}^n$. Then $u^T v = \langle u, v \rangle$.

Proof. Let $u = \begin{bmatrix} u_1 \\ \vdots \\ u_n \end{bmatrix}$ and $v = \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix}$.

Then $u^T v = \begin{bmatrix} u_1 & \cdots & u_n \end{bmatrix} \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix} = u_1 v_1 + u_2 v_2 + \cdots + u_n v_n$

$= \langle u, v \rangle$.

Theorem 6 in Section 6.2.

$P \in M_{n \times n}$ is orthogonal $\Leftrightarrow$ The columns of P form an orthonormal basis for $\mathbb{R}^n$

Sketch of proof.

P orthogonal $\iff P^T P = I$

$$\iff \begin{bmatrix} u_1^T \\ \vdots \\ u_n^T \end{bmatrix} \begin{bmatrix} u_1 & \cdots & u_n \end{bmatrix} = \begin{bmatrix} 1 & & 0 \\ & \ddots & \\ 0 & & 1 \end{bmatrix}$$

where $P = [u_1 \cdots u_n]$ has columns $u_1, \ldots, u_n \in \mathbb{R}^n$.

$$\iff \begin{bmatrix} u_1^T u_1 & u_1^T u_2 & \cdots & u_1^T u_n \\ \vdots & & & \\ u_n^T u_1 & \cdots & & u_n^T u_n \end{bmatrix} = \begin{bmatrix} 1 & & 0 \\ & \ddots & \\ 0 & & 1 \end{bmatrix}$$

$$\iff \langle u_j, u_k \rangle = u_j^T u_k = \begin{cases} 1 & j = k \\ 0 & j \neq k. \end{cases}$$

$\iff \{u_1, \ldots, u_n\}$ is an orthonormal basis for $\mathbb{R}^n$

Warning. If a matrix has orthogonal columns which are <u>not</u> orthonormal then the matrix is <u>not</u> orthogonal!!

Example

$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ is orthogonal since

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}^T \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

Also the columns $\left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$ form an orthonormal basis for $\mathbb{R}^2$.

Similarly $I \in M_{n \times n}$ is orthogonal.

Theorem

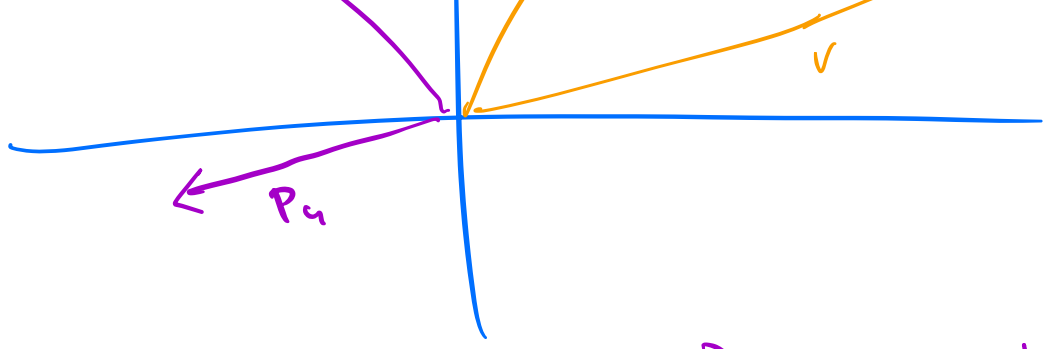
$P \in M_{n \times n}$ is orthogonal $\iff \langle Pu, Pv \rangle = \langle u, v \rangle$ $\forall\, u, v \in \mathbb{R}^n$.

(In particular multiplication by P preserves norm and angle.)

Sketch of proof

$$\begin{aligned} \langle Pu, Pv \rangle &= (Pu)^T Pv \\ &= u^T P^T P v \\ &= u^T I v \qquad \text{if } P \text{ is orthogonal.} \\ &= u^T v \\ &= \langle u, v \rangle. \end{aligned}$$

Example



P is a rotation $\Rightarrow$ P is orthogonal!

Reflections also produce orthogonal matrices

Def. A matrix $A \in M_{n\times n}$ is <u>orthogonally diagonalizable</u> if there is an orthogonal matrix $P \in M_{n\times n}$ and a diagonal matrix $D \in M_{n\times n}$ such that

$$A = PDP^{-1}. \qquad (*)$$

Suppose $(*)$ holds and write

$P = [u_1 \cdots u_n]$, where $u_1, \dots, u_n \in \mathbb{R}^n$

are the columns of P. By Thm 6 §6.2 $\{u_1, \dots, u_n\}$ is an orthonormal basis for $\mathbb{R}^n$.

From long ago, we also know that $u_1, \dots, u_n$ are eigenvectors of A (Thm 5 §5.3).

In fact if $D = \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix}$ then

$$A u_1 = \lambda_1 u_1, \quad \ldots, \quad A u_n = \lambda_n .$$

This proves

Theorem

A is orthogonally diagonalizable

$\iff$ there is an orthonormal basis for $\mathbb{R}^n$ consisting of eigenvectors of A.

Example 2 in textbook.

Take $A = \begin{bmatrix} 6 & -2 & -1 \\ -2 & 6 & -1 \\ -1 & -1 & 5 \end{bmatrix}$, a symmetric matrix.

$$\det(A - xI) = \cdots = -(x-8)(x-6)(x-3)$$

$\Rightarrow$ eigenvalues are $8, 6,$ and 3.

$\Rightarrow$ A diagonalizable (3 distinct eigenvalues for $\mathbb{R}^3$)

After some computation,

$$\mathrm{Nul}(A - 8I) = \mathrm{Span}\left\{ \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} \right\}$$

$$\mathrm{Nul}(A - 6I) = \mathrm{Span}\left\{ \begin{bmatrix} -1 \\ -1 \\ 2 \end{bmatrix} \right\}$$

$$\mathrm{Nul}(A - 3I) = \mathrm{Span}\left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right\}.$$

$$\Rightarrow \left\{ \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ -1 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right\}$$ basis of eigenvectors.

Check that this is an orthogonal basis.